# Determine Distance on the Coordinate Plane

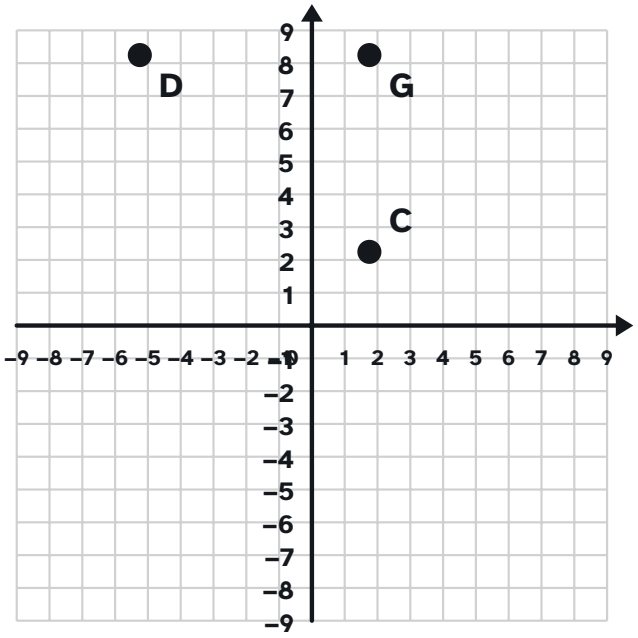
Find two distances on the coordinate plane and compare them.

**Name** _____ **Date** _____

## Tonight's spark

A park map has a hub at G, (1.75, 8.25). Cruz walks straight to C, (1.75, 2.25). Dara walks straight to D, (–5.25, 8.25). One unit is one block. Who walks farther?

**Guess first: Cruz goes down and Dara goes left. Who walks farther?**




**Our class model: find the shared coordinate, then use the other**

## Try it

Work it out on paper. Write one answer on each line.

**1** **Which coordinate do (3, 1) and (3, –5) share?**

**A.** The x-coordinate **B.** The y-coordinate **C.** Neither one

**My answer** _____

**2** **Find the distance between (2, 6) and (2, 1).**

**A.** 7 **B.** 5 **C.** 6 **D.** 1

**My answer** _____

**3** **Find the distance between (–4, 3) and (5, 3).**

**A.** 1 **B.** 4 **C.** 5 **D.** 9

**My answer** _____

**4** **Find the distance between (1.5, 4) and (1.5, –2.5).**

**A.** 6.5 **B.** 1.5 **C.** 4 **D.** 2.5

**My answer** _____

**5** **Ana walks from (0, 5) to (0, –3). Beto walks from (0, 5) to (6, 5). Who walks farther?**

**A.** Beto, by 4 units **B.** Ana, by 2 units **C.** Ana, by 14 units **D.** They walk the same

**My answer** _____

### Talk about it

Which coordinate does each trip share? Do you add or subtract for each trip, and who goes farther?

### Home mission

Cross two forks on the table to make the axes. Put one coin 3 finger-widths above the crossing and another 5 below it. Ask your student: how far apart are the coins? Add or subtract? Why?

### If we get stuck, ask

- Which coordinate do these two points share?
- Do the other two values have the same sign or opposite signs?
- Does the trip cross an axis? Then what do we do?

**Do it on a screen**

Scan or type the link. Same lesson, with hints.

**eduwonderlab.com/lessons/7-6-part2/homework.html**

**How did it feel? (optional)**

Tricky Getting there Got it